

SALES DATA SQL QUERIES

1. Business Understanding

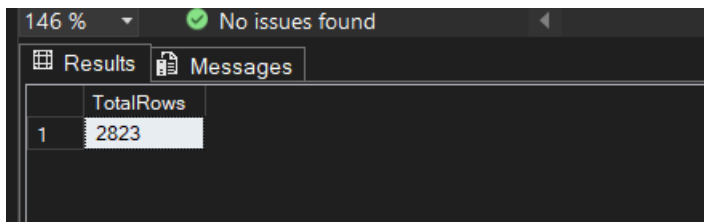
Goal:

- Measure revenue performance
- Identify top customers
- Analyze trends
- Support business decisions

2. Data Exploration (EDA in SQL)

a) Total Records

```
--Total Records--  
SELECT COUNT(*) AS TotalRows  
FROM sales_data_sample;
```



	TotalRows
1	2823

b) Distinct Countries

```
-- Distinct Countries --  
  
SELECT DISTINCT Country  
FROM sales_data_sample;
```

146 % No issues found

Results		Messages
	Country	
1	Finland	
2	USA	
3	Italy	
4	Philippines	
5	Germany	
6	Switzerland	
7	Australia	
8	Sweden	
9	Austria	
10	UK	
11	Canada	
12	Ireland	
13	Norway	
14	France	
15	Belgium	
16	Japan	
17	Spain	

3. Data Cleaning

a) Remove Duplicates

```
-- Remove Duplicates --  
  
SELECT OrderNumber, COUNT(*)  
FROM sales_data_sample  
GROUP BY OrderNumber  
HAVING COUNT(*) > 1;
```

146 % No issues found

Results Messages

	OrderNumber	(No column name)
1	10411	9
2	10388	8
3	10125	2
4	10102	2
5	10248	14
6	10171	4
7	10271	11
8	10148	14
9	10265	2
10	10365	3
11	10219	4
12	10156	2
13	10357	10
14	10394	7
15	10348	8
16	10311	11
17	10302	6

Query executed successfully.

B)Fix Date Format

```
--Fix Date Format--

UPDATE sales_data_sample
SET OrderDate = TRY_CONVERT(DATETIME, OrderDate);
```

4. Core Business Metrics

a)Total Revenue

```
--Total Revenue--

SELECT SUM(Sales) AS TotalRevenue
FROM sales_data_sample;
```

146 % ✓ No issues found

Results Messages

	TotalRevenue
1	10032628.8493042

b)Total Orders

```
--Total Orders--  
  
SELECT COUNT(DISTINCT OrderNumber) AS TotalOrders  
FROM sales_data_sample;
```

146 % ✓ No issues found

Results Messages

	TotalOrders
1	307

c)Total Customers

```
--Total Customers--  
  
SELECT COUNT(DISTINCT CustomerName) AS TotalCustomers  
FROM sales_data_sample;
```

146 % ✓ No issues found

Results Messages

	TotalCustomers
1	92

5. Trend Analysis

a) Monthly Revenue Trend

```
--Monthly Revenue Trend--  
  
✓ SELECT  
    Year_ID,  
    Month_ID,  
    SUM(Sales) AS MonthlyRevenue  
FROM sales_data_sample  
GROUP BY Year_ID, Month_ID  
ORDER BY Year_ID, Month_ID;
```

146 % No issues found

	Year_ID	Month_ID	MonthlyRevenue
1	2003	1	129753.599731445
2	2003	2	140836.189941406
3	2003	3	174504.900085449
4	2003	4	201609.551086426
5	2003	5	192673.109985352
6	2003	6	168082.56072998
7	2003	7	187731.880493164
8	2003	8	197809.300048828
9	2003	9	263973.360961914
10	2003	10	568290.971557617
11	2003	11	1029837.66271973
12	2003	12	261876.459899902
13	2004	1	316577.418762207
14	2004	2	311419.530029297
15	2004	3	205733.729858398
16	2004	4	206148.119750977
17	2004	5	273438.389648438

b) Quarterly Performance

--Quarterly Performance--

```
SELECT
    Year_ID,
    QTR_ID,
    SUM(Sales) AS QuarterlyRevenue
FROM sales_data_sample
GROUP BY Year_ID, QTR_ID
ORDER BY Year_ID, QTR_ID;
```

146 % No issues found

	Year_ID	QTR_ID	QuarterlyRevenue
1	2003	1	445094.689758301
2	2003	2	562365.221801758
3	2003	3	649514.541503906
4	2003	4	1860005.09417725
5	2004	1	833730.678649902
6	2004	2	766260.730529785
7	2004	3	1109396.26745605
8	2004	4	2014774.91674805
9	2005	1	1071992.35809326
10	2005	2	719494.350585938

6. Regional Analysis

a) Revenue by Country

--Revenue by Country--

```
SELECT
    Country,
    SUM(Sales) AS Revenue
FROM sales_data_sample
GROUP BY Country
ORDER BY Revenue DESC;
```

46 % No issues found

Results Messages

	Country	Revenue
1	USA	3627982.82574463
2	Spain	1215686.92236328
3	France	1110916.52178955
4	Australia	630623.098754883
5	UK	478880.45892334
6	Italy	374674.310974121
7	Finland	329581.910339355
8	Norway	307463.699707031
9	Singapore	288488.41027832
10	Denmark	245637.150634766
11	Canada	224078.559936523
12	Germany	220472.08972168
13	Sweden	210014.210205078
14	Austria	202062.530334473
15	Japan	188167.81060791
16	Switzerland	117713.55859375
17	Belgium	108412.619628906

b) Top 3 Countries

```
--Top 3 Countries--
SELECT TOP 3 Country, SUM(Sales) AS Revenue
FROM sales_data_sample
GROUP BY Country
ORDER BY Revenue DESC;
```

146 % No issues found

Results Messages

	Country	Revenue
1	USA	3627982.82574463
2	Spain	1215686.92236328
3	France	1110916.52178955

7. Customer Analysis

a) Top 5 Customers

```
-- Top 5 Customers --
```

```
SELECT TOP 5
    CustomerName,
    SUM(Sales) AS TotalSpent
FROM sales_data_sample
GROUP BY CustomerName
ORDER BY TotalSpent DESC;
```

140 % NO issues found

	CustomerName	TotalSpent
1	Euro Shopping Channel	912294.110473633
2	Mini Gifts Distributors Ltd.	654858.058105469
3	Australian Collectors, Co.	200995.41015625
4	Muscle Machine Inc	197736.940185547
5	La Rochelle Gifts	180124.899719238

b) Customer Ranking

```
--Customer Ranking --
```

```
SELECT
    CustomerName,
    SUM(Sales) AS TotalSales,
    RANK() OVER (ORDER BY SUM(Sales) DESC) AS Rank
FROM sales_data_sample
GROUP BY CustomerName;
```


146 % ✓ No issues found

Results Messages

	CustomerName	TotalSales	Rank
1	Euro Shopping Channel	912294.110473633	1
2	Mini Gifts Distributors Ltd.	654858.058105469	2
3	Australian Collectors, Co.	200995.41015625	3
4	Muscle Machine Inc	197736.940185547	4
5	La Rochelle Gifts	180124.899719238	5
6	Dragon Souvenirs, Ltd.	172989.680541992	6
7	Land of Toys Inc.	164069.439331055	7
8	The Sharp Gifts Warehouse	160010.270263672	8
9	AV Stores, Co.	157807.809631348	9
10	Anna's Decorations, Ltd	153996.129150391	10
11	Souvenirs And Things Co.	151570.979858398	11
12	Corporate Gift Ideas Co.	149882.500244141	12
13	Salzburg Collectables	149798.630187988	13
14	Danish Wholesale Imports	145041.600830078	14
15	Saveley & Henriot, Co.	142874.25	15
16	L'ordine Souvenirs	142601.330932617	16
17	Rovelli Gifts	137955.720153809	17

8. Product Analysis

a) Sales by Product Line

```
--Sales by Product Line--

SELECT
    ProductLine,
    SUM(Sales) AS Revenue
FROM sales_data_sample
GROUP BY ProductLine
ORDER BY Revenue DESC;
```

146 % No issues found

Results Messages

	ProductLine	Revenue
1	Classic Cars	3919615.6607666
2	Vintage Cars	1903150.83557129
3	Motorcycles	1166388.3392334
4	Trucks and Buses	1127789.84326172
5	Planes	975003.571350098
6	Ships	714437.130126953
7	Trains	226243.468994141

9. Deal Size Analysis

b) Revenue by Deal Size

```
--Revenue by Deal Size--

SELECT
    DealSize,
    SUM(Sales) AS Revenue
FROM sales_data_sample
GROUP BY DealSize
ORDER BY Revenue DESC;
```

146 % No issues found

Results Messages

	DealSize	Revenue
1	Medium	6087432.23974609
2	Small	2643077.34881592
3	Large	1302119.26074219

c) Running Total Revenue

--Running Total Revenue--

```
SELECT
    OrderDate,
    SUM(Sales) OVER (ORDER BY OrderDate) AS RunningRevenue
FROM sales_data_sample;
```

146 % No issues found

	OrderDate	RunningRevenue
1	2003-01-06 00:00:00.0000000	12133.25
2	2003-01-06 00:00:00.0000000	12133.25
3	2003-01-06 00:00:00.0000000	12133.25
4	2003-01-06 00:00:00.0000000	12133.25
5	2003-01-09 00:00:00.0000000	23565.58984375
6	2003-01-09 00:00:00.0000000	23565.58984375
7	2003-01-09 00:00:00.0000000	23565.58984375
8	2003-01-09 00:00:00.0000000	23565.58984375
9	2003-01-10 00:00:00.0000000	30429.6398925781
10	2003-01-10 00:00:00.0000000	30429.6398925781
11	2003-01-29 00:00:00.0000000	85131.6396484375
12	2003-01-29 00:00:00.0000000	85131.6396484375
13	2003-01-29 00:00:00.0000000	85131.6396484375
14	2003-01-29 00:00:00.0000000	85131.6396484375
15	2003-01-29 00:00:00.0000000	85131.6396484375
16	2003-01-29 00:00:00.0000000	85131.6396484375
17	2003-01-29 00:00:00.0000000	85131.6396484375

Query executed successfully. SUBHADIP

d) Moving Average (3 Months)

--Moving Average (3 Months)--

```
SELECT
    Year_ID,
    Month_ID,
    AVG(Sales) OVER (
        ORDER BY Year_ID, Month_ID
        ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
    ) AS MovingAvg
FROM sales_data_sample;
```

146 % No issues found

Results Messages

	Year_ID	Month_ID	MovingAvg
1	2003	1	5404.6201171875
2	2003	1	5401.43994140625
3	2003	1	4732.61995442708
4	2003	1	4917.24658203125
5	2003	1	4656.419921875
6	2003	1	4831.42659505208
7	2003	1	4448.02994791667
8	2003	1	3594.68334960938
9	2003	1	4005.01668294271
10	2003	1	3554.88663736979
11	2003	1	3999.63997395833
12	2003	1	3540.17667643229
13	2003	1	3648.20336914063
14	2003	1	2914.86336263021
15	2003	1	2915.11995442708
16	2003	1	2402.08325195313
17	2003	1	2675.78991699219

Query executed successfully. SUBHADIP\SQLEXPRESS

11. Business Insights (Say This Clearly)

- Revenue is concentrated among **top customers**
- Certain countries dominate **sales performance**
- Sales show **seasonal peaks**
- Specific product lines generate **maximum revenue**
- Large deals significantly impact revenue.